ModelIt!

Student Activity Pack

Can We Actually Live on Mars?

Modeling Survival in the Most Hostile Environment Humans Have Ever Faced

Grade 09L1 | NGSS HS-ESS1-2, HS-ETS1-2 | Systems and System Models

STUDENT NAME _____ DATE _____

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Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you think are the biggest challenges to keeping humans alive on Mars?

2 How is Mars different from Earth in terms of what humans need to survive?

3 Draw what you think a Mars habitat would need to include to keep six people alive for a year.

4 Could we grow food on Mars? What would that require?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Atmospheric Pressure | Radiation Exposure | Water Availability | Temperature | Crop Viability

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Optimal Habitat

All systems functioning at ideal levels

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Pressure Breach

Atmospheric Pressure drops to near-Mars surface levels

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Water Crisis

Water Availability drops to critical levels

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Energy Production — The power generation capacity of the habitat using solar panels or nuclear reactors to run life support, heating, and agricultural systems

Oxygen Generation — The production of breathable oxygen through electrolysis of water or CO2 processing from the Martian atmosphere

Waste Recycling Efficiency — How effectively the habitat converts human waste, CO2, and wastewater back into usable resources in a closed-loop system

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

What would it actually take to keep humans alive on Mars — and is it even possible? Use evidence from your simulation to support your answers.

1. **Why is Mars colonization often described as 'the hardest engineering problem in human history'?**

2. **What does your model reveal about the difference between surviving on Mars and thriving on Mars?**

3. **Which of the five components is the most critical — and what happens if it fails?**

4. **How does your model change when you add energy production or oxygen generation?**

5. **What assumptions does your model make that might not hold true on the actual Martian surface?**

8 STEM Challenge: Design a Mars Habitat Module

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YOUR ENGINEERING MISSION

Design a habitat module that addresses the five critical survival factors and can sustain a crew of six for one year using available Martian resources.

THE SCENARIO

NASA has announced the Mars Colony Design Challenge for high school students. Your team must design a habitat module that solves pressure, radiation, water, temperature, and food production simultaneously. The module must use in-situ Martian resources wherever possible and include failure contingency plans.

Engineering Design Thinking

1. WHAT SHAPE AND MATERIALS WILL PROVIDE THE BEST PRESSURE CONTAINMENT AND RADIATION SHIELDING?
2. HOW WILL YOU EXTRACT WATER FROM MARTIAN ICE AND RECYCLE WASTEWATER?
3. WHAT CROPS ARE BEST SUITED FOR MARS HABITAT AGRICULTURE AND HOW MUCH GROWING SPACE DO SIX PEOPLE NEED?
4. WHAT BACKUP SYSTEMS WILL YOU INCLUDE FOR EACH CRITICAL FAILURE MODE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Aerospace Engineers design spacecraft, habitats, and life support systems for space missions. They work at NASA, SpaceX, Blue Origin, and defense contractors, earning $80,000–$160,000/year. Astrobiologists who study the potential for life on other planets earn $60,000–$130,000/year.